

Linear Functions & Function Notation

SUPPORTING SKILLS LEGEND

Student learning objectives for this unit

I — Introduce

D — Deepen

A — Apply

R — Reinforce

- ☐ 1. I can **interpret** function notation as a statement about input and output.

RELATED SKILLS

- I Use function notation to express and solve an input-output question.
- D Identify input and output values in a formula, graph, table, or context.
- A Identify the input and output represented by function notation.

- ☐ 2. I can **determine** whether a graph or table represents a function.

RELATED SKILLS

- D Apply the vertical line test to a graph.
- D Identify input and output values in a formula, graph, table, or context.

- ☐ 3. I can **evaluate** a function or solve for an input using a graph, table, or formula.

RELATED SKILLS

- D Identify input and output values in a formula, graph, table, or context.
- D Solve for an input associated with a specified output.
- A Use function notation to express and solve an input-output question.
- A Use inverse operations to maintain an equivalent equation.

- ☐ 4. I can **identify** independent and dependent variables in a relationship.

RELATED SKILLS

- I Identify independent and dependent variables.
- A Identify input and output values in a formula, graph, table, or context.

- ☐ 5. I can **calculate** average rate of change from two points.

RELATED SKILLS

- D Calculate average rate of change over an interval.
- A Calculate slope from two points.

- ☐ 6. I can **interpret** interval notation and identify increasing or decreasing intervals.

RELATED SKILLS

- I Describe increasing or decreasing behavior over intervals.
- D Translate between inequality and interval notation.

☐ **7. I can **determine** total change from an average rate of change over an interval.**

RELATED SKILLS

- ☐ I Use average rate of change and interval length to calculate total change.
 - ☐ A Add, subtract, multiply, and divide signed integers.
 - ☐ A Calculate average rate of change over an interval.
-

☐ **8. I can **determine and interpret** the units of an average rate of change.**

RELATED SKILLS

- ☐ I Determine rate units from input- and output-axis units.
 - ☐ D Attach and interpret units for rates, inputs, and outputs.
-

☐ **9. I can **identify** a linear function from a table, graph, or formula.**

RELATED SKILLS

- ☐ I Use constant differences or ratios to distinguish linear and exponential patterns.
 - ☐ D Connect parameters in a formula to features of its graph.
 - ☐ D Use tabular patterns to predict graph shape and behavior.
-

☐ **10. I can **interpret** the slope and intercept of a linear model.**

RELATED SKILLS

- ☐ D Explain a model parameter in the language of its context.
 - ☐ A Attach and interpret units for rates, inputs, and outputs.
 - ☐ A Determine slope from a graph.
 - ☐ A Plot an intercept accurately on a coordinate plane.
-

☐ **11. I can **determine** the intersection of two linear relationships graphically or algebraically.**

RELATED SKILLS

- ☐ A Add equations to eliminate a variable.
 - ☐ A Check an ordered pair in both equations of a system.
 - ☐ A Locate and interpret the intersection of two lines.
 - ☐ A Substitute an equivalent expression for a variable.
-

☐ **12. I can **write** equations of horizontal, vertical, parallel, and perpendicular lines through a specified point.**

RELATED SKILLS

- ☐ D Determine the negative reciprocal of a nonzero slope.
 - ☐ D Recognize that distinct parallel lines have equal slopes.
 - ☐ A Convert among common forms of a linear equation.
 - ☐ A Translate quantities and relationships into an equation.
-